

This fictional city brief was created for teaching RAG retrieval and evaluation.

This fictional glossary supports comparison across the Rivergate, Sunford, Lakemont, and Northbridge adaptation briefs. It defines recurring terms that students may encounter when evaluating retrieval results. The glossary is not a complete climate textbook. It is a compact reference for the exercise.

Climate adaptation means changing systems so they can handle climate impacts that are already happening or likely to happen. Flood barriers, cooling centers, green roofs, and backup bus routes are adaptation measures. Climate mitigation means reducing the causes of climate change, for example by lowering greenhouse gas emissions.

Heat mitigation and heat response are related but different. Heat mitigation reduces heat before an emergency, such as shade trees, cool roofs, and reflective surfaces. Heat response protects people during an active heat event, such as cooling centers, water distribution, and wellness checks.

Nature-based solutions use natural processes to reduce risk. Rivergate's wetland restoration, Sunford's shade trees, and Lakemont's rain gardens are examples. Nature-based solutions can provide extra benefits such as cleaner water, habitat, and cooler streets, but they still require maintenance and monitoring.

Social vulnerability describes how climate risk is shaped by age, income, disability, health, housing, language access, and transport options. A flood warning is less useful if a resident cannot leave without help. A heat alert is less useful if a worker cannot change schedule or reach a cooling center.

Resilience does not mean that nothing fails. In these briefs, resilience means that failures are less likely, warnings arrive earlier, damage is smaller, and recovery is faster. Northbridge may still lose tram service during a severe storm, but backup route B-12 can keep essential trips moving.

Evaluation questions should usually name an expected source. A flood question is often answered by Rivergate. A heat-health question is often answered by Sunford. A green roof or permeable pavement question is often answered by Lakemont. A transport disruption question is often answered by Northbridge. A definition question is often answered by this glossary.

Good RAG answers should stay grounded in retrieved context. If the context does not contain the answer, the system should say that it does not know based on the provided documents. This matters because confident unsupported answers are harder to detect than simple retrieval failures.